Singular Value Decomposition (SVD)

August 27, 2025

Definition of Singular Values

Core Definition

Let A be an $m \times n$ matrix. A scalar σ is called a **singular value** of A if there exists a pair of non-zero vectors $\mathbf{u}$ (in $\mathbb{R}^m$) and $\mathbf{v}$ (in $\mathbb{R}^n$) such that:

$$A\mathbf{v} = \sigma\mathbf{u}$$
$$A^T\mathbf{u} = \sigma\mathbf{v}$$

- The vector $\mathbf{u}$ is called a **left singular vector**.
- The vector $\mathbf{v}$ is called a **right singular vector**.

A More Practical Definition

Connection to Eigenvalues

The singular values of a matrix A are the positive square roots of the eigenvalues of $A^T A$ (or AA^T). If $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of $A^T A$, then the singular values are:

$$\sigma_i = \sqrt{\lambda_i}$$

Key Properties

1. The matrices $A^T A$ and AA^T have the same set of non-zero eigenvalues.
2. The **left singular vectors** ($\mathbf{u}_i$) are the eigenvectors of AA^T.
3. The **right singular vectors** ($\mathbf{v}_i$) are the eigenvectors of $A^T A$.

Quick Example: Finding Singular Values

Example

Let $A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix}$. Since A is 2×3, it's easier to compute the smaller matrix AA^T (2×2).

$$AA^T = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 6 & 12 \end{pmatrix}$$

The eigenvalues of AA^T are $\lambda_1 = 15$ and $\lambda_2 = 0$. Therefore, the singular values of A are:

$$\sigma_1 = \sqrt{15} \quad \text{and} \quad \sigma_2 = \sqrt{0} = 0$$

Singular Value Decomposition (SVD)

Theorem (Singular Value Decomposition)

Let A be a matrix of order $m \times n$ with non-zero singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$.
Then A can be factored as:

$$A = U\Sigma V^T$$

where:

- U *is an $m \times m$ orthogonal matrix whose columns are the* **left singular vectors** *of A (eigenvectors of AA^T).*
- V *is an $n \times n$ orthogonal matrix whose columns are the* **right singular vectors** *of A (eigenvectors of $A^T A$).*
- Σ *is an $m \times n$ diagonal matrix containing the singular values.*

$$\Sigma = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix} \quad \text{where} \quad D = diag(\sigma_1, \ldots, \sigma_r)$$

Working Procedure for SVD (for $m \geq n$)

1. **Find V:** Determine an orthonormal set of eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ of the $n \times n$ matrix $A^T A$. The matrix V has these vectors as its columns.

$$V = \begin{pmatrix} | & & | \\ \mathbf{v}_1 & \cdots & \mathbf{v}_n \\ | & & | \end{pmatrix}$$

2. **Find Σ:** The singular values $\sigma_i = \sqrt{\lambda_i}$ are the square roots of the eigenvalues of $A^T A$. Form the $m \times n$ matrix Σ.
3. **Find U:** For the non-zero singular values σ_i (where $i = 1, \ldots, r$), calculate the first r columns of U:

$$\mathbf{u}_i = \frac{1}{\sigma_i} A \mathbf{v}_i$$

   If needed, find the remaining $m - r$ orthonormal vectors $\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m$ that are orthogonal to $\mathbf{u}_1, \ldots, \mathbf{u}_r$.
4. **Assemble:** The SVD is $A = U \Sigma V^T$.

Problem 1: SVD of a 2×3 Matrix

Find the SVD of $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix}$

Step 1: Compute $A^T A$ and its eigenvalues.

$$A^T A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 2 \end{pmatrix}$$

The eigenvalues are $\lambda_1 = 4, \lambda_2 = 2, \lambda_3 = 0$. The singular values are $\sigma_1 = 2, \sigma_2 = \sqrt{2}, \sigma_3 = 0$.

Step 2: Find the matrix V. The normalized eigenvectors of A^TA are:

$$\lambda_1 = 4 \implies \mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$\lambda_2 = 2 \implies \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\lambda_3 = 0 \implies \mathbf{v}_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

So, $V = \begin{pmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \end{pmatrix}$.

Problem 1: SVD (Continued)

Find the SVD of $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix}$

Step 3: Find the matrix U. We use $\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i$ for the non-zero singular values.
For $\sigma_1 = 2$:

$$\mathbf{u}_1 = \frac{1}{2} A\mathbf{v}_1 = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For $\sigma_2 = \sqrt{2}$:

$$\mathbf{u}_2 = \frac{1}{\sqrt{2}} A\mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Since U must be 2×2, we have all its columns: $U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.

Problem 1: Final Decomposition

Final Assembly: $A = U\Sigma V^T$

We have:

- $U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
- $\Sigma = \begin{pmatrix} 2 & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix}$ (an $m \times n = 2 \times 3$ matrix)
- $V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{pmatrix}$

So the SVD of A is:

$$A = \underbrace{\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} 2 & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix}}_{\Sigma} \underbrace{\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{pmatrix}^T}_{V^T}$$

Problem 2: A Permutation Matrix

Find the SVD of $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

Step 1: Compute $A^T A$.

$$A^T A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

The eigenvalues are $\lambda_1 = 4, \lambda_2 = 1, \lambda_3 = 1$. The singular values are $\sigma_1 = 2, \sigma_2 = 1, \sigma_3 = 1$.
Step 2: Find V. The eigenvectors of the diagonal matrix $A^T A$ are the standard basis vectors:

$$\mathbf{v}_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

So, $V = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$.

Problem 2: Final Decomposition

Find the SVD of $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

Step 3: Find U.

$$\mathbf{u}_1 = \frac{1}{2} A\mathbf{v}_1 = \frac{1}{2} \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\mathbf{u}_2 = \frac{1}{1} A\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\mathbf{u}_3 = \frac{1}{1} A\mathbf{v}_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

So, $U = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$.

Final Assembly:

$$A = \underbrace{\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}^T}_{V^T}$$

Problem 3

Find the SVD of $A = \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}$

Step 1: Compute A^TA and its eigenvalues.

$$A^TA = \begin{bmatrix} -3 & 6 & 6 \\ 1 & -2 & -2 \end{bmatrix} \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix} = \begin{bmatrix} 81 & -27 \\ -27 & 9 \end{bmatrix}$$

The eigenvalues are $\lambda_1 = 90, \lambda_2 = 0$. The singular values are $\sigma_1 = 3\sqrt{10}, \sigma_2 = 0$.

Step 2: Find the matrix V and Σ. The normalized eigenvectors of A^TA are:

$$\lambda_1 = 90 \implies \mathbf{v}_1 = \frac{1}{\sqrt{10}} \begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

$$\lambda_2 = 0 \implies \mathbf{v}_2 = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

So, $V = \frac{1}{\sqrt{10}} \begin{pmatrix} -3 & 1 \\ 1 & 3 \end{pmatrix}$. Note that $\Sigma = \begin{pmatrix} 3\sqrt{10} & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$

Problem 3: Final Decomposition

Step 3: Find U.

$$\mathbf{u}_1 = \frac{1}{3\sqrt{10}} A\mathbf{v}_1 = \frac{1}{3}\begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix}$$

We need two more columns for U : $\mathbf{u}_2, \mathbf{u}_3$, which must be orthonormal to $\mathbf{u}_1$ and to each other. They form a basis for the null space of A^T. Solving $A^T\mathbf{y} = 0$ gives $-3y_1 + 6y_2 + 6y_3 = 0 \implies y_1 = 2y_2 + 2y_3$. We need an orthonormal basis for it. Let's pick two independent vectors in the plane: If $y_2 = 1, y_3 = 0$, then $y_1 = 2$. Vector $\mathbf{w}_2 = (2, 1, 0)^T$. If $y_2 = 0, y_3 = 1$, then $y_1 = 2$. Vector $\mathbf{w}_3 = (2, 0, 1)^T$. Now use Gram-Schmidt to find an orthonormal basis $\{\mathbf{u}_2, \mathbf{u}_3\}$.

- $\mathbf{u}_2 = \frac{\mathbf{w}_2}{\|\mathbf{w}_2\|} = \frac{1}{\sqrt{5}}(2, 1, 0)^T$.
- $\mathbf{u}_3 = \frac{1}{3\sqrt{5}}(2, -4, 5)^T$.

$$U = \begin{pmatrix} 1/3 & 2/\sqrt{5} & 2/(3\sqrt{5}) \\ -2/3 & 1/\sqrt{5} & -4/(3\sqrt{5}) \\ -2/3 & 0 & 5/(3\sqrt{5}) \end{pmatrix}$$

Final Decomposition: $A = \begin{pmatrix} 1/3 & 2/\sqrt{5} & 2/(3\sqrt{5}) \\ -2/3 & 1/\sqrt{5} & -4/(3\sqrt{5}) \\ -2/3 & 0 & 5/(3\sqrt{5}) \end{pmatrix} \begin{pmatrix} 3\sqrt{10} & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \left(\frac{1}{\sqrt{10}} \begin{pmatrix} -3 & 1 \\ 1 & 3 \end{pmatrix} \right)^T.$

Problem 3 (Method 2)

Find the singular value decomposition (SVD) of the matrix A:

$$A = \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}$$

Step 1: Calculate AA^T

We start by computing AA^T. Since A is 3×2, AA^T will be a 3×3 matrix.

$$AA^T = \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix} \begin{bmatrix} -3 & 6 & 6 \\ 1 & -2 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} (-3)^2 + 1^2 & (-3)(6) + (1)(-2) & (-3)(6) + (1)(-2) \\ (6)(-3) + (-2)(1) & 6^2 + (-2)^2 & 6^2 + (-2)^2 \\ (6)(-3) + (-2)(1) & 6^2 + (-2)^2 & 6^2 + (-2)^2 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & -20 & -20 \\ -20 & 40 & 40 \\ -20 & 40 & 40 \end{bmatrix}$$

Step 2: Find Eigenvalues of AA^T

We need to find the roots of the characteristic equation $\det(AA^T - \lambda I) = 0$.

- **Observation:** The second and third rows of AA^T are identical, and the first row is a multiple of the others ($R_2 = -2R_1$). This means the matrix has rank 1.
- An $n \times n$ matrix of rank r has $n - r$ zero eigenvalues.
- Here, $n = 3$ and $r = 1$, so we must have $3 - 1 = 2$ zero eigenvalues.
- The sum of eigenvalues equals the trace of the matrix: $\text{Tr}(AA^T) = 10 + 40 + 40 = 90$.
- So, $\lambda_1 + \lambda_2 + \lambda_3 = 90$. Since two eigenvalues are 0, the third must be 90.

Eigenvalues

The eigenvalues of AA^T are $\lambda_1 = 90$, $\lambda_2 = 0$, $\lambda_3 = 0$.

Step 3: Find Singular Values and Construct Σ

The singular values σ_i are the square roots of the positive eigenvalues, ordered from largest to smallest.

$$\sigma_i = \sqrt{\lambda_i}$$

- $\sigma_1 = \sqrt{90} = \sqrt{9 \times 10} = 3\sqrt{10}$
- The rank of matrix A is $r = 1$, corresponding to the one non-zero singular value.

The matrix Σ has the same dimensions as A (3×2).

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 3\sqrt{10} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Step 4: Find Left Singular Vectors (Columns of U)

The columns of U are the orthonormal eigenvectors of AA^T.
For $\lambda_1 = 90$: Solve $(AA^T - 90I)\mathbf{x} = \mathbf{0}$.

$$\begin{bmatrix} -80 & -20 & -20 \\ -20 & -50 & 40 \\ -20 & 40 & -50 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0}$$

This system reduces to $x = \frac{1}{2}y$ and $y = -z$. This gives an eigenspace defined by $\frac{x}{1} = \frac{y}{-2} = \frac{z}{-2}$. An eigenvector is $\begin{pmatrix} 1, -2, -2 \end{pmatrix}^T$.
Normalize it: $\| \begin{pmatrix} 1 & -2 & -2 \end{pmatrix}^T \| = \sqrt{1^2 + (-2)^2 + (-2)^2} = \sqrt{9} = 3$.

$$\mathbf{u_1} = \frac{1}{3} \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix}$$

Step 4: Find Left Singular Vectors (continued)

For $\lambda_{2,3} = 0$: Solve $(AA^T - 0I)\mathbf{x} = \mathbf{0}$.

$$\begin{bmatrix} 10 & -20 & -20 \\ -20 & 40 & 40 \\ -20 & 40 & 40 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0} \implies x - 2y - 2z = 0$$

We need an orthonormal basis for this plane. This can be found using the Gram-Schmidt process on a basis like $\left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \right\}$.
The orthonormal basis vectors are:

$$\mathbf{u_2} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{u_3} = \frac{1}{3\sqrt{5}} \begin{bmatrix} 2 \\ 5 \\ -4 \end{bmatrix}$$

Assembling U

We combine these orthonormal vectors as columns to form U.

$$U = \begin{bmatrix} \mathbf{u_1} & \mathbf{u_2} & \mathbf{u_3} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{2}{\sqrt{5}} & \frac{2}{3\sqrt{5}} \\ -\frac{2}{3} & 0 & \frac{5}{3\sqrt{5}} \\ -\frac{2}{3} & \frac{1}{\sqrt{5}} & -\frac{4}{3\sqrt{5}} \end{bmatrix}$$

Step 5: Find Right Singular Vectors (Columns of V)

We use the relationship $\mathbf{v_i} = \frac{1}{\sigma_i} A^T \mathbf{u_i}$ for non-zero σ_i.

For $\sigma_1 = 3\sqrt{10}$:

$$\mathbf{v_1} = \frac{1}{3\sqrt{10}} A^T \mathbf{u_1} = \frac{1}{3\sqrt{10}} \begin{bmatrix} -3 & 6 & 6 \\ 1 & -2 & -2 \end{bmatrix} \left(\frac{1}{3} \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} \right)$$

$$= \frac{1}{9\sqrt{10}} \begin{bmatrix} -3 - 12 - 12 \\ 1 + 4 + 4 \end{bmatrix} = \frac{1}{9\sqrt{10}} \begin{bmatrix} -27 \\ 9 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

For $\mathbf{v_2}$, we need a unit vector orthogonal to $\mathbf{v_1}$. For a vector $\begin{pmatrix} a \\ b \end{pmatrix}$, an orthogonal vector is $\begin{pmatrix} -b \\ a \end{pmatrix}$. An orthogonal vector is $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$ or $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$. Let's use $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$.

Normalize it: $\| \begin{pmatrix} 1 & 3 \end{pmatrix}^T \| = \sqrt{1^2 + 3^2} = \sqrt{10}$.

$$\mathbf{v_2} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Step 6: Construct the Matrix V and V^T

We form V using $\mathbf{v_1}$ and $\mathbf{v_2}$ as columns.

$$V = \begin{bmatrix} \mathbf{v_1} & \mathbf{v_2} \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} -3 & 1 \\ 1 & 3 \end{bmatrix}$$

Then, we find its transpose for the final formula.

$$V^T = \left(\frac{1}{\sqrt{10}} \begin{bmatrix} -3 & 1 \\ 1 & 3 \end{bmatrix} \right)^T = \frac{1}{\sqrt{10}} \begin{bmatrix} -3 & 1 \\ 1 & 3 \end{bmatrix}$$

(Note that in this specific case, V is symmetric, so $V = V^T$).

Final Result: The Singular Value Decomposition

Combining our results, we have the SVD: $A = U\Sigma V^T$.

$$\underbrace{\begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} \frac{1}{3} & \frac{2}{\sqrt{5}} & \frac{2}{3\sqrt{5}} \\ -\frac{2}{3} & 0 & \frac{5}{3\sqrt{5}} \\ -\frac{2}{3} & \frac{1}{\sqrt{5}} & -\frac{4}{3\sqrt{5}} \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} 3\sqrt{10} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}}_{\Sigma} \underbrace{\frac{1}{\sqrt{10}} \begin{bmatrix} -3 & 1 \\ 1 & 3 \end{bmatrix}}_{V^T}$$

Earlier approach: $A = \begin{pmatrix} 1/3 & 2/\sqrt{5} & 2/(3\sqrt{5}) \\ -2/3 & 1/\sqrt{5} & -4/(3\sqrt{5}) \\ -2/3 & 0 & 5/(3\sqrt{5}) \end{pmatrix} \begin{pmatrix} 3\sqrt{10} & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \left(\frac{1}{\sqrt{10}} \begin{pmatrix} -3 & 1 \\ 1 & 3 \end{pmatrix} \right)^T$.